

Block Ciphers and AES

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References and notes used from:

Cryptography and Network Security, Principles and Practices, by William Stallings
(combination of several editions)

Cryptography and network security course at SCSE, NTU by Prof. Anwitaman Datta

Recall DES

What is DES encryption?

DES is a symmetric block cipher (shared secret key), with a key length of 56-bits. Published as the Federal Information Processing Standards (FIPS) 46 standard in 1977, DES was officially withdrawn in 2005.

The federal government originally developed DES encryption over 35 years ago to provide cryptographic security for all government communications. The idea was to ensure government systems all used the same, secure standard to facilitate interconnectivity.

DES was broken

To show that the DES was inadequate and should not be used in important systems anymore, a series of challenges were sponsored to see how long it would take to decrypt a message. Two organizations played key roles in breaking DES: distributed.net and the Electronic Frontier Foundation (EFF).

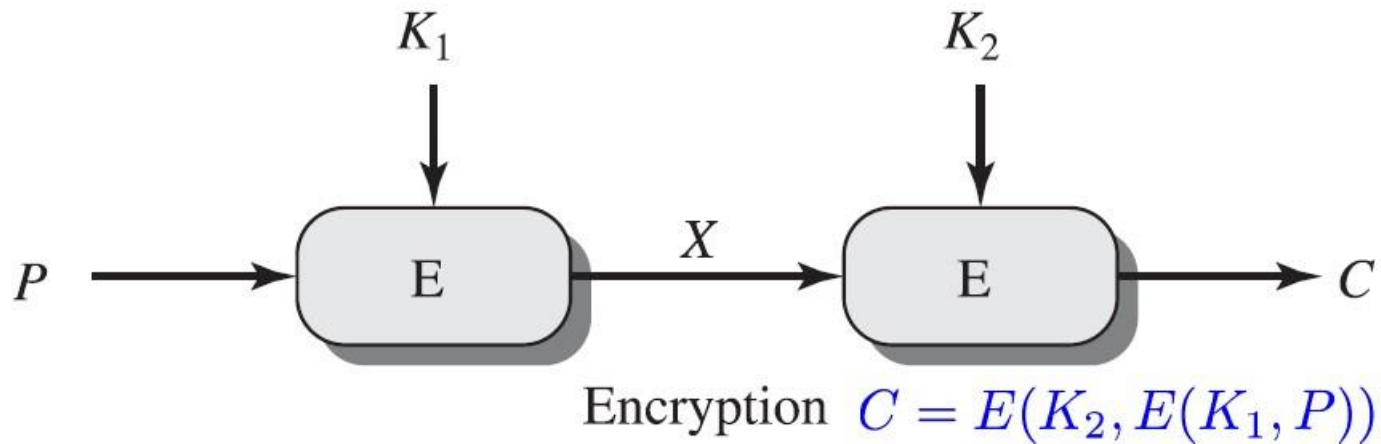
DES was broken

- The **DES I contest** (1997) took 84 days to break the encrypted message using a brute force attack.
- In 1998, there were **two DES II challenges** issued. The first challenge took just over a month and the decrypted text was *“The unknown message is: Many hands make light work”*. The second challenge took less than three days, with the plaintext message *“It’s time for those 128-, 192-, and 256-bit keys”*.
- The final DES III challenge in early 1999 only took 22 hours and 15 minutes. Electronic Frontier Foundation’s Deep Crack computer (built for less than \$250,000) and distributed.net’s computing network found the 56-bit DES key, deciphered the message, and they (EFF & distributed.net) won the contest. The decrypted message read *“See you in Rome (Second AES Candidate Conference, March 22-23, 1999)”*, and was found after checking about 30 percent of the key space – finally proving that DES belonged to the past.

What to do with a broken cipher?

In 1998, Electronic Frontier Foundation breaks by using a \$250K machine

Not a problem!! Apply DES twice



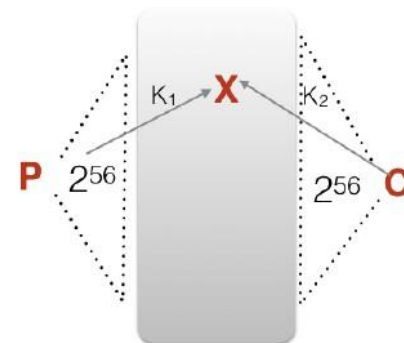
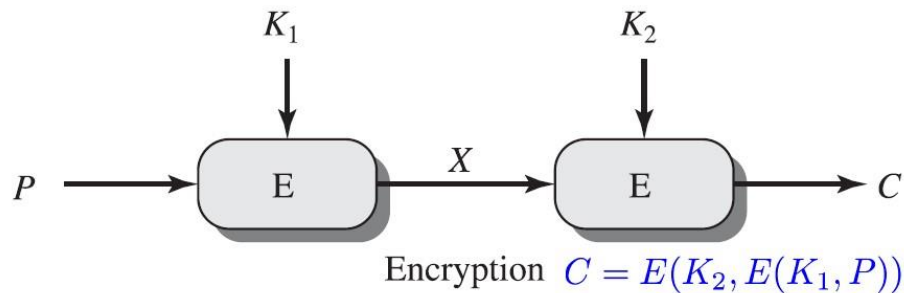
What to do with a broken cipher?

What if

1. Cascaded block cipher is equivalent to one cipher?

What if: $E(K_2, E(K_1, P)) \equiv E(K_3, P)$

2. Meet-in-the-middle attack – Use known plain text and also break the second DES



What to do with a broken cipher?

1. Encrypt three times!! – This is called Triple DES
2. Meet-in-the-middle attack – very low threat probability
3. Use two keys for three rounds – that's good enough

$$C = E(K_1, D(K_2, E(K_1, P)))$$

$$P = D(K_1, E(K_2, D(K_1, P)))$$

What's the problem of triple DES?

1. Computation efficiency is a problem
2. NIST released a call for proposal for Advanced Encryption Standard after the DES development
3. Roughly 15 proposals were received
4. After shortlist and review, Rijndael was accepted.
5. Introduced to standards in 2001



Vincent Rijmen
born in 1970



Joan Daemen
born in 1965

AES

Evaluation criteria by NIST

1. General security
2. Software implementation
3. Restricted-space environment
4. Hardware implementation
5. Attacks on implementation
6. Encryption and Decryption
7. Key agility

AES

Rijndael was designed to have the following characteristics:

- Resistance against all known attacks
- Speed and code compactness on a wide range of platforms
- Design simplicity

Key size (words/bytes/bits)	4/16/128	6/24/192	8/32/256
Plaintext block size (words/bytes/bits)	4/16/128	4/16/128	4/16/128
Number of rounds	10	12	14
Round key size (words/bytes/bits)	4/16/128	4/16/128	4/16/128
Expanded key size (words/bytes)	44/176	52/208	60/240

Important changes w.r.t. DES

All operations done on strings of bytes (8 bits)

1 byte is viewed as an element in a finite field of size 256 (using polynomial ring)

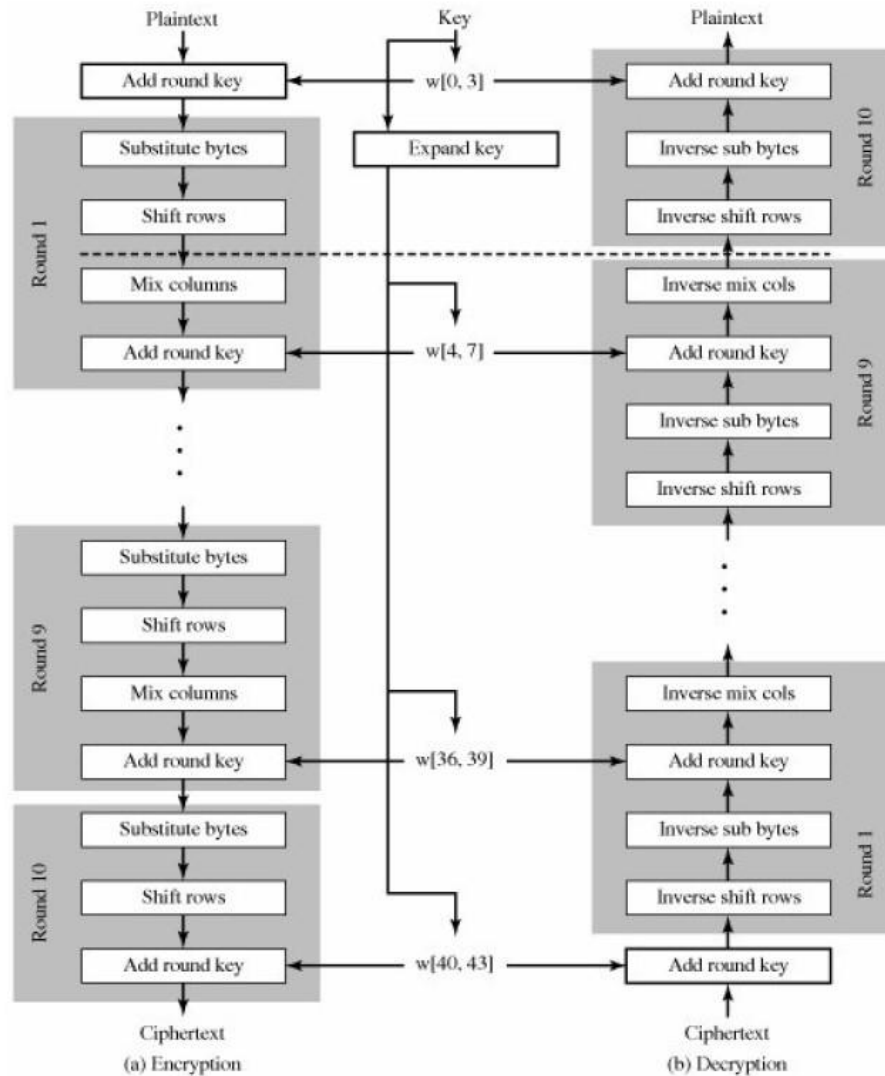
Irreducible polynomial must be of degree? [8 4 3 1 0]

Encryption algorithm involves addition, multiplication and division

irreducible polynomial

$$x^8 + x^4 + x^3 + x + 1$$

AES – Big Picture



AES – How to arrange the plaintext and key?

Input bits of size 128 bits – 16 bytes

Write them in the form of a 4 X 4 matrix

in_0	in_4	in_8	in_{12}
in_1	in_5	in_9	in_{13}
in_2	in_6	in_{10}	in_{14}
in_3	in_7	in_{11}	in_{15}

Key is of size 128 bits – 16 bytes

Write them also in the form of a 4 X 4 matrix

k_0	k_4	k_8	k_{12}
k_1	k_5	k_9	k_{13}
k_2	k_6	k_{10}	k_{14}
k_3	k_7	k_{11}	k_{15}

AES

There are a total of 10 rounds

Each round has 4 functions to perform

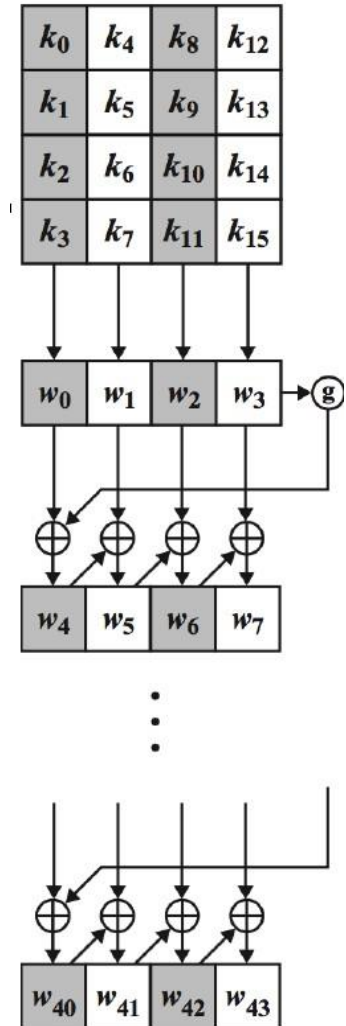
1. Substitution
2. ShiftRows
3. MixColumns
4. AddRoundKey

Each round needs a 128 bit key (somewhat like Feistel structure)

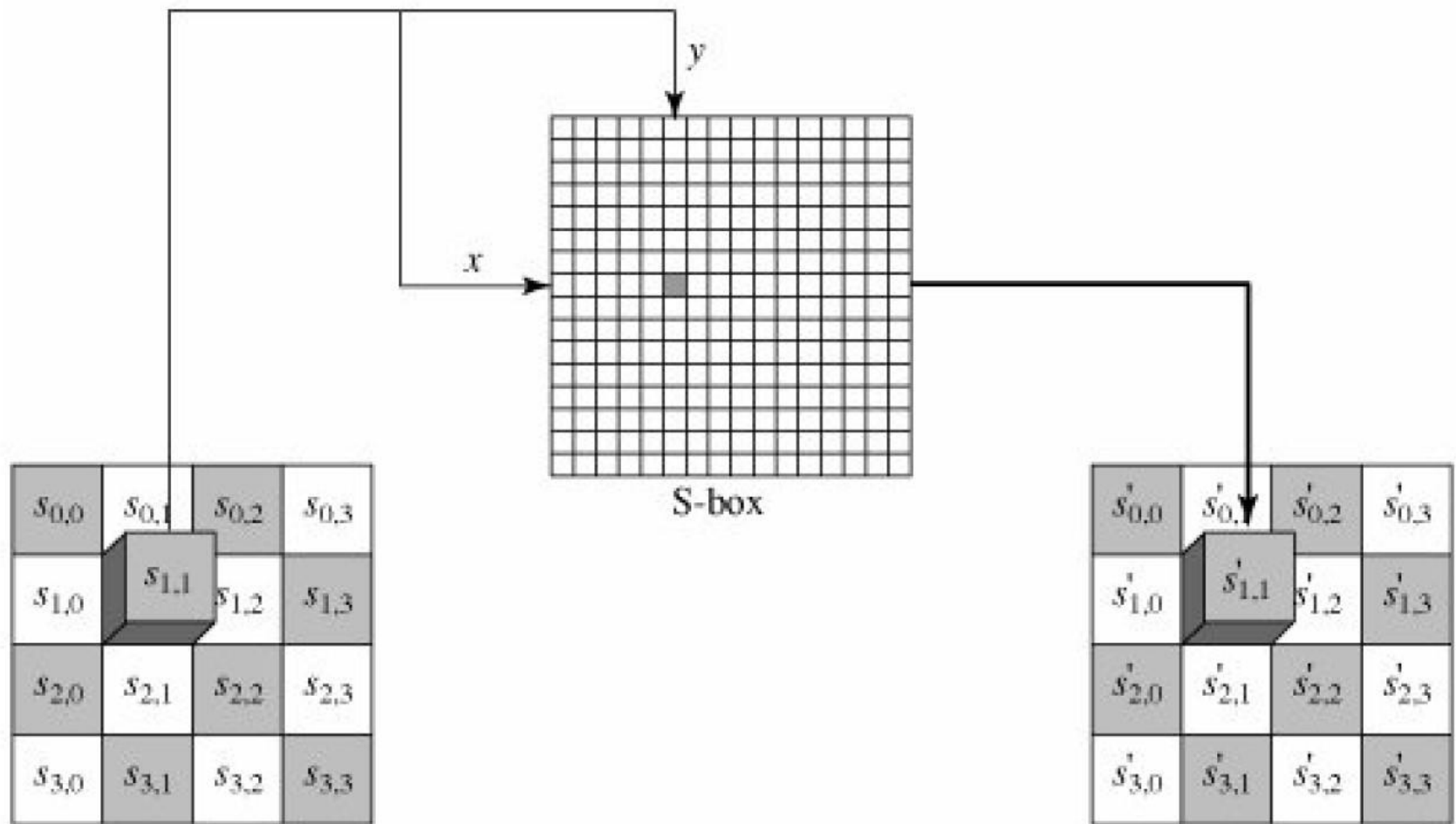
How to generate key sequences for 10 rounds using 128 bits initial key?

AES – Key Expansion

1. Use the 128 bit key to generate 10 more 128 bits one for each round
2. Write the 128 bit key as four words. Word = 4 bytes
3. Generate 4 more words for each round using a g function
 1. g function is specific to the round



AES – Substitution Block



AES – Substitution Block

		<i>y</i>															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
<i>x</i>	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

AES – Inside the Substitution Block

What happens to 95?

95 → 2A as per S-box

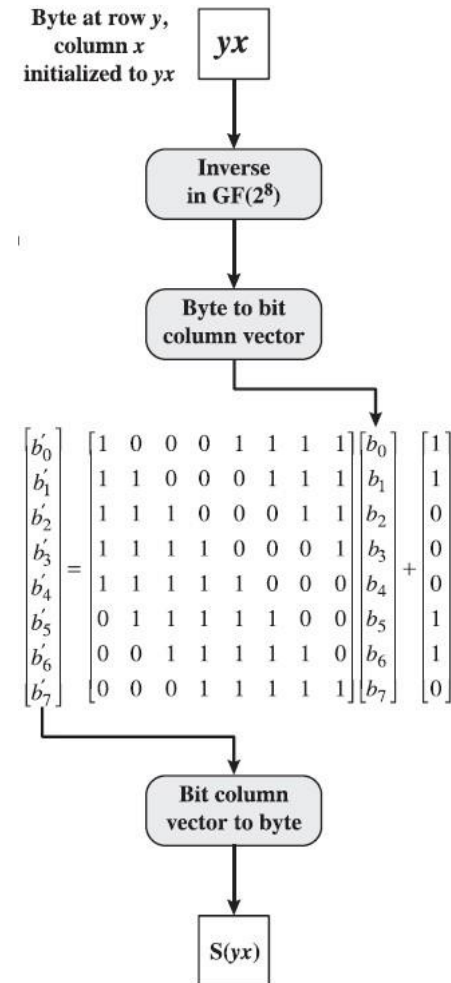
Write 95 as binary

Its inverse is 8A in the field

8A = [1 0 0 0 1 0 1 0]

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \oplus \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \oplus \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Answer is 2A

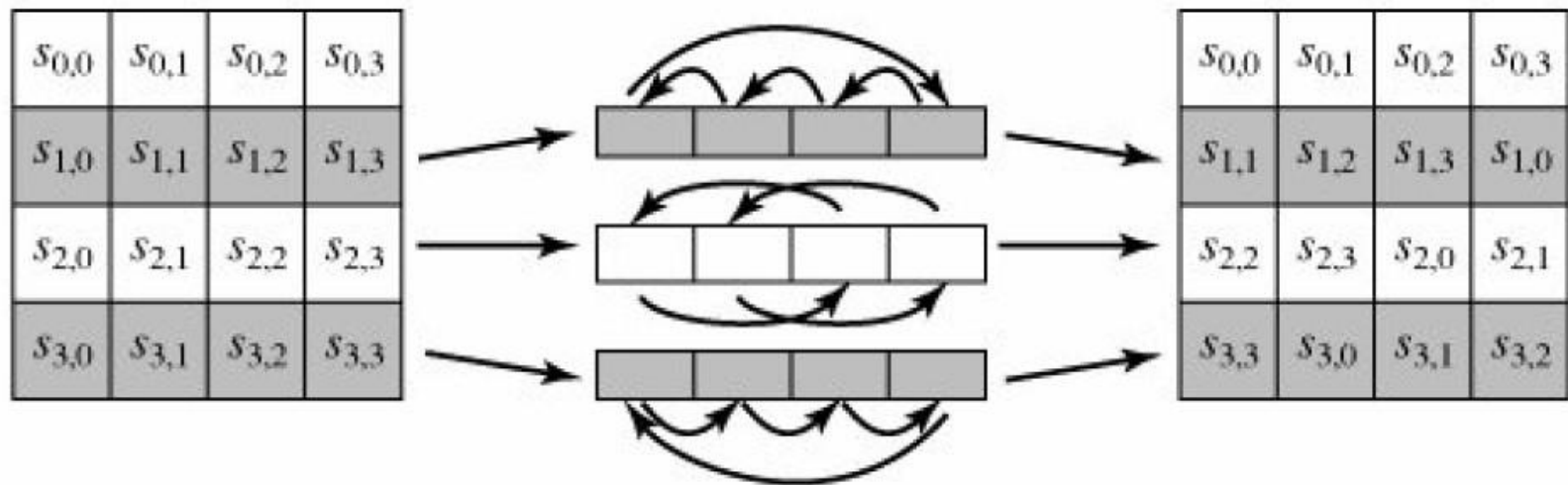


(a) Calculation of byte at row y , column x of S-box

AES – Inside the Substitution Block

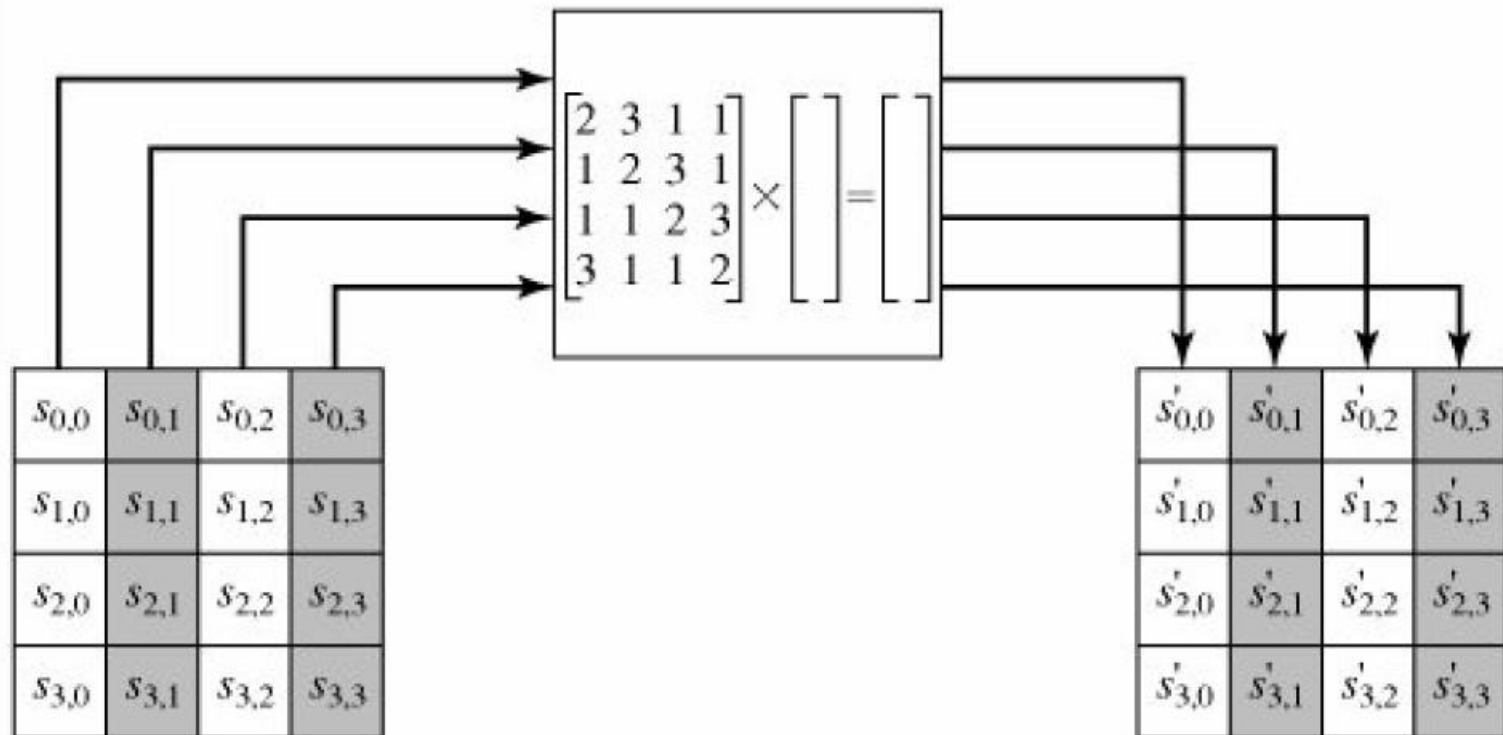
- Low correlation between input/output bits
- Output a **non-linear** function of input
using multiplicative inverse provides non-linearity
- **Constant** chosen so that:
there are no fixed points: $S(a) \neq a$
there are no opposite fixed points: $S(a) \neq \bar{a}$
 $\bar{a}$ is the bit-wise complement of a
- S-box is **invertible**, but there are **no self-inverses**
 $S(a) \neq S^{-1}(a)$

AES - ShiftRows



- i^{th} row gets $i-1$ left circular shift
4 bytes of a column are spread to four different columns

AES - MixColumn



AES - MixColumn

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix}$$

$$s'_{0,j} = (2 \cdot s_{0,j}) \oplus (3 \cdot s_{1,j}) \oplus s_{2,j} \oplus s_{3,j}$$

$$s'_{1,j} = s_{0,j} \oplus (2 \cdot s_{1,j}) \oplus (3 \cdot s_{2,j}) \oplus s_{3,j}$$

$$s'_{2,j} = s_{0,j} \oplus s_{1,j} \oplus (2 \cdot s_{2,j}) \oplus (3 \cdot s_{3,j})$$

$$s'_{3,j} = (3 \cdot s_{0,j}) \oplus s_{1,j} \oplus s_{2,j} \oplus (2 \cdot s_{3,j})$$

AES – MixColumn (Example)

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

→

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

$$\begin{aligned}
 (\{02\} \cdot \{87\}) \oplus (\{03\} \cdot \{6E\}) \oplus \{46\} \oplus \{A6\} &= \{47\} \\
 \{87\} \oplus (\{02\} \cdot \{6E\}) \oplus (\{03\} \cdot \{46\}) \oplus \{A6\} &= \{37\} \\
 \{87\} \oplus \{6E\} \oplus (\{02\} \cdot \{46\}) \oplus (\{03\} \cdot \{A6\}) &= \{94\} \\
 (\{03\} \cdot \{87\}) \oplus \{6E\} \oplus \{46\} \oplus (\{02\} \cdot \{A6\}) &= \{ED\}
 \end{aligned}$$

For the first equation, we have $\{02\} \cdot \{87\} = (0000\ 1110) \oplus (0001\ 1011) = (0001\ 0101)$; and $\{03\} \cdot \{6E\} = \{6E\} \oplus (\{02\} \cdot \{6E\}) = (0110\ 1110) \oplus (1101\ 1100) = (1011\ 0010)$. Then

$$\begin{aligned}
 \{02\} \cdot \{87\} &= 0001\ 0101 \\
 \{03\} \cdot \{6E\} &= 1011\ 0010 \\
 \{46\} &= 0100\ 0110 \\
 \{A6\} &= 1010\ 0110 \\
 0100\ 0111 &= \{47\}
 \end{aligned}$$

AES - AddRoundKey

This is byte-by-byte XOR

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

 $\oplus$

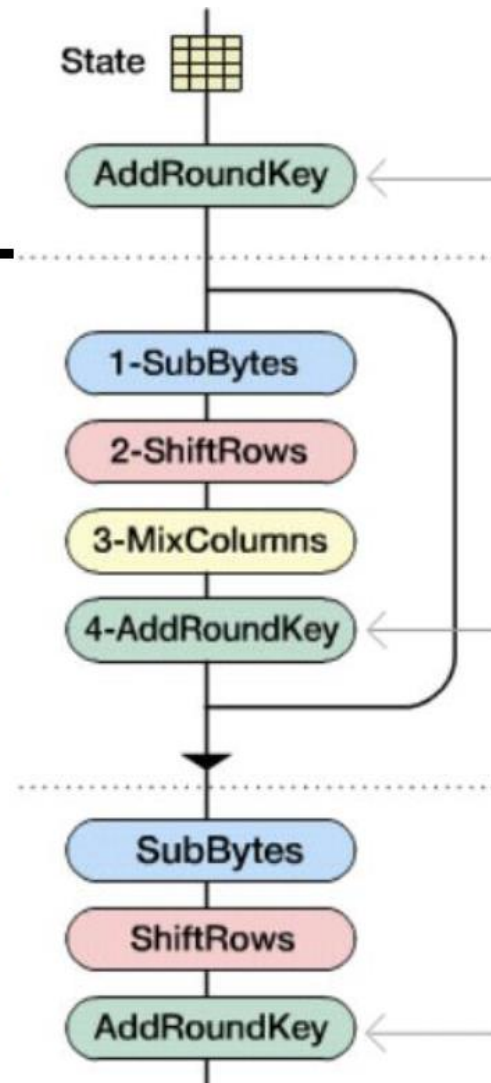
AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

 $=$

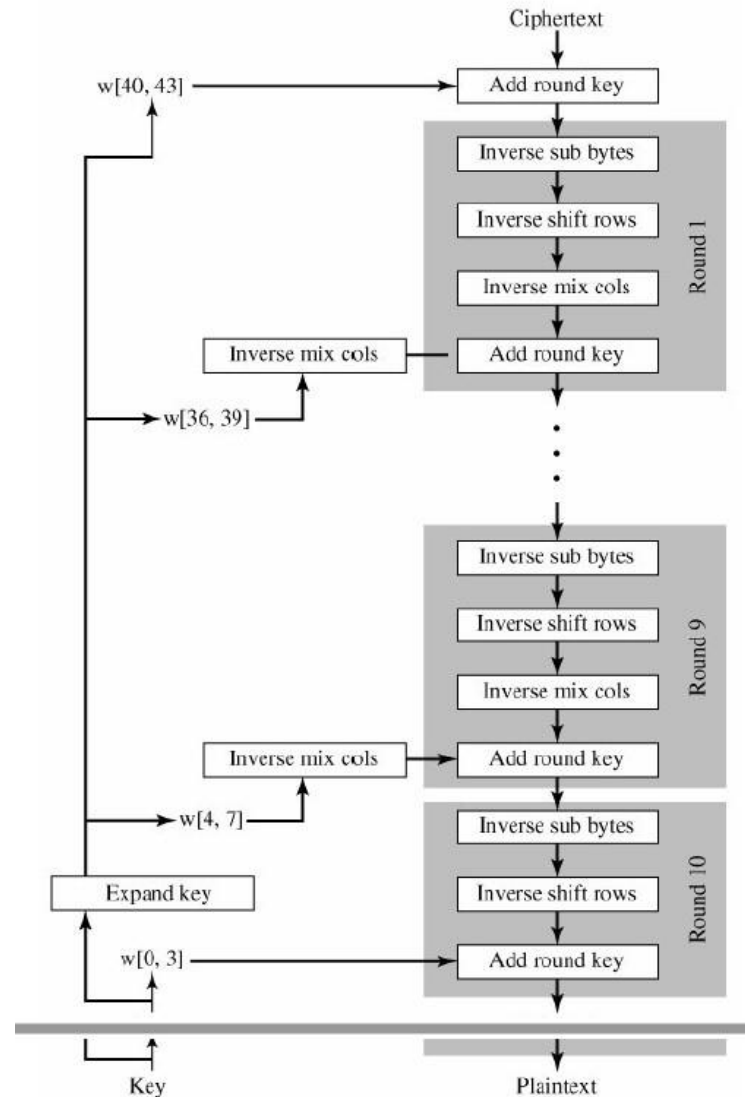
EB	59	8B	1B
40	2E	A1	C3
F2	38	13	42
1E	84	E7	D2

AES: Wrap-up

- The cipher begins and ends with an AddRoundKey stage. **Why?**
It's in effect a Vernam (one-time pad) cipher
- The other three stages together provide confusion, diffusion and non-linearity, but by themselves, they provide no security.
Why? No secrets involved in the other steps

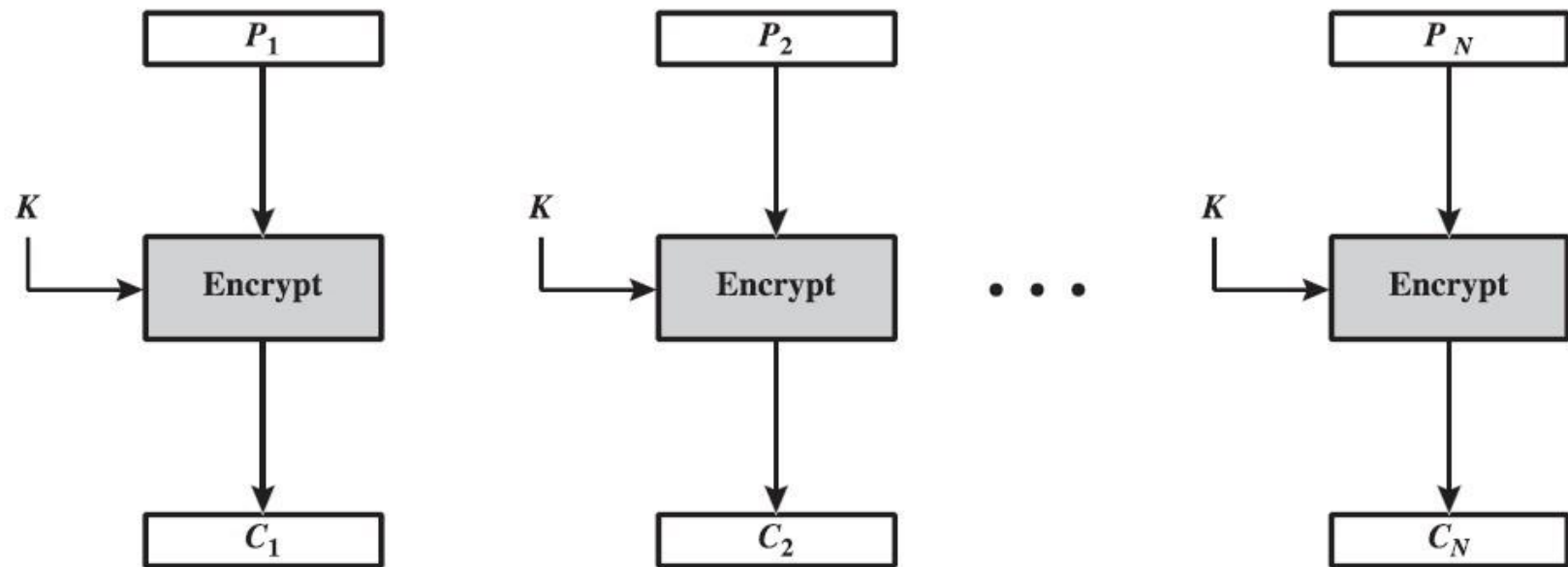


AES – Decryption Algorithm



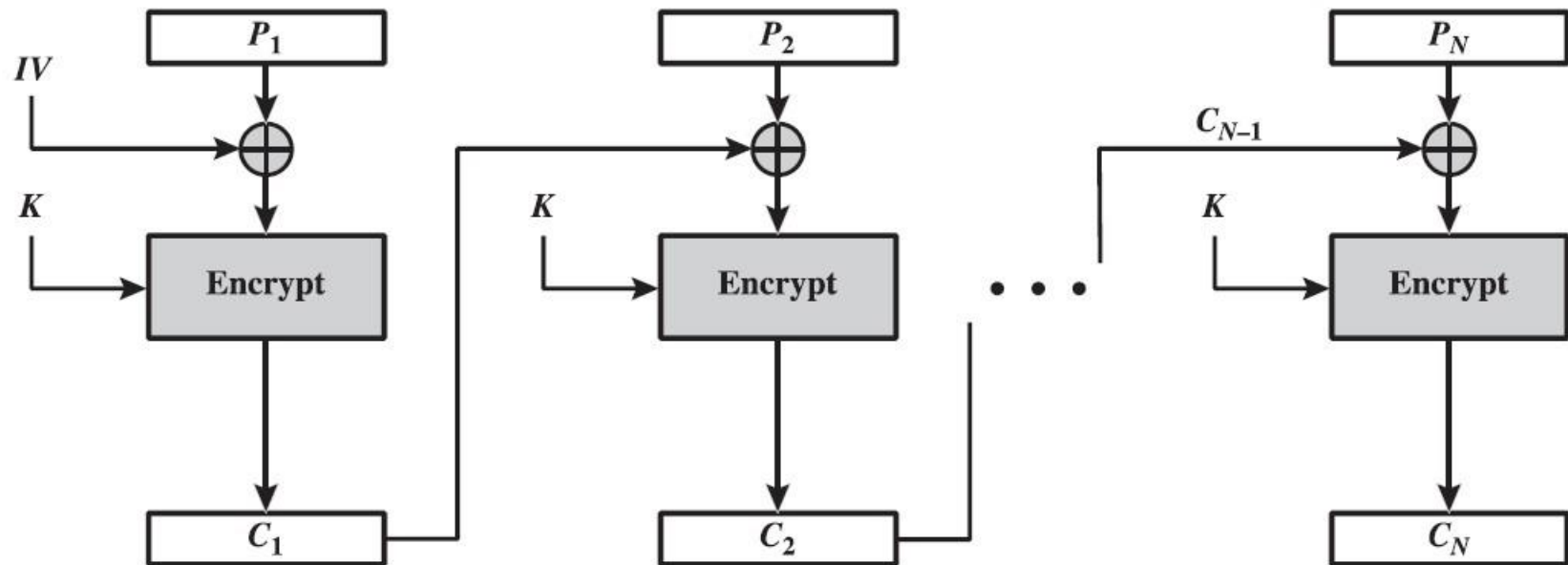
Block Ciphers – What is the problem?

Plain text longer than the key



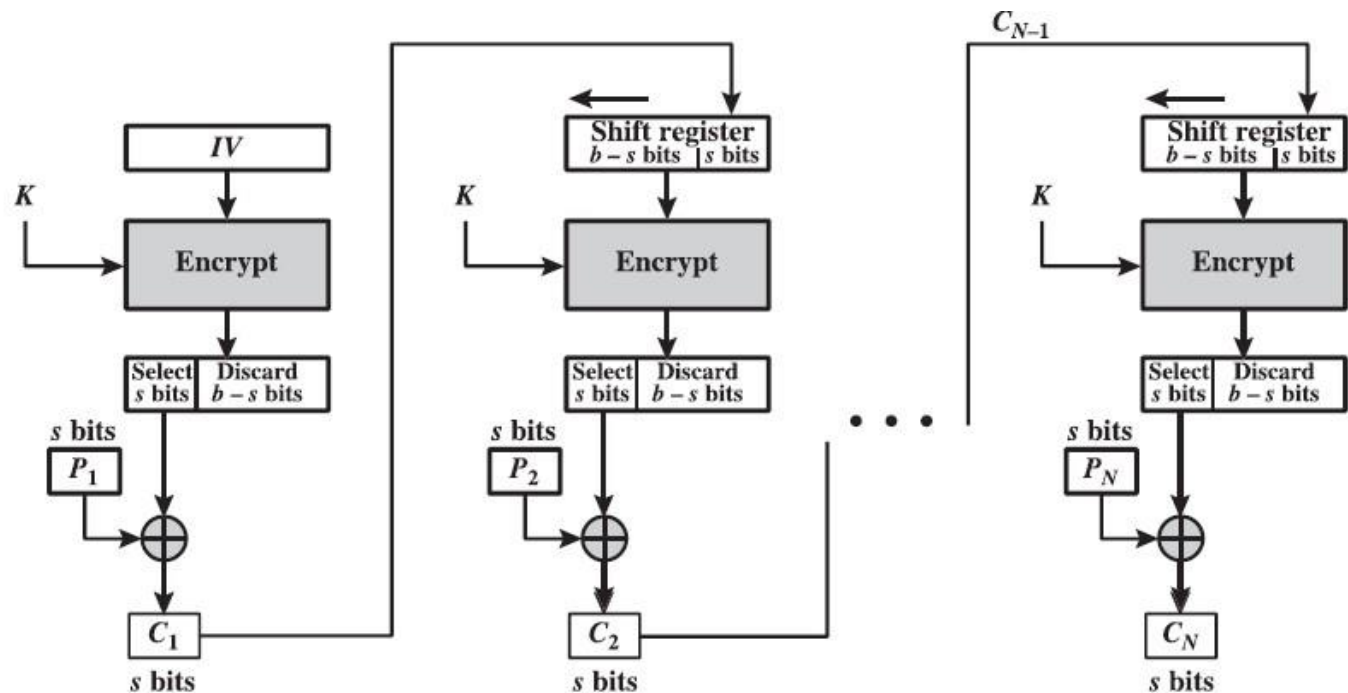
Block Ciphers – Link them block by block

Plain text longer than the key – Chain them to avoid replicas



Stream Cipher using Block Ciphers

Plain text longer than the key – Chain them to avoid replicas but....



More on AES and Block Ciphers

Recommend more reading on this from Stallings